ICT305 Topics in IT

Topic 4
Generative AI

CIHE Australia

Materials

- References
  - *Introduction to generative AI, University of Wincosin*
  - Janice Kung, *"ChatGPT 101: Introduction to Generative AI"*, University of Alberta,

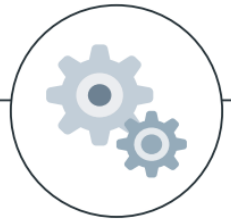
Agenda

- **Recognize** different types of generative AI tools and understand how this technology is continuously evolving.
- **Explain** the basics of what generative AI tools like ChatGPT can do, including their strengths and limitations.
- **Safely and responsibly use** a variety of generative AI tools in different environments.

Generative AI: Using Gen AI to Talk About Gen AI

- Dall-E-3. Prompt: “Create an image that illustrates Generative AI at Crown Institute of Higher education”
- Please try this prompt in your Generative AI tool (e.g., ChatGPT) and compare the output with your friends' ouput.

Evolution of Artificial Intelligence



Machine Learning

Stage-1
Artificial Narrow Intelligence

Specialises in one area to solve a specific problem.

E.g. Siri, Alexa, Google Assistant

Machine Intelligence

Stage-2
Artificial General Intelligence

Generalized to solve problems as well as any human.

Machine Consciousness

Stage-3
Artificial Super Intelligence

Intellect much smarter than the best humans in all fields.

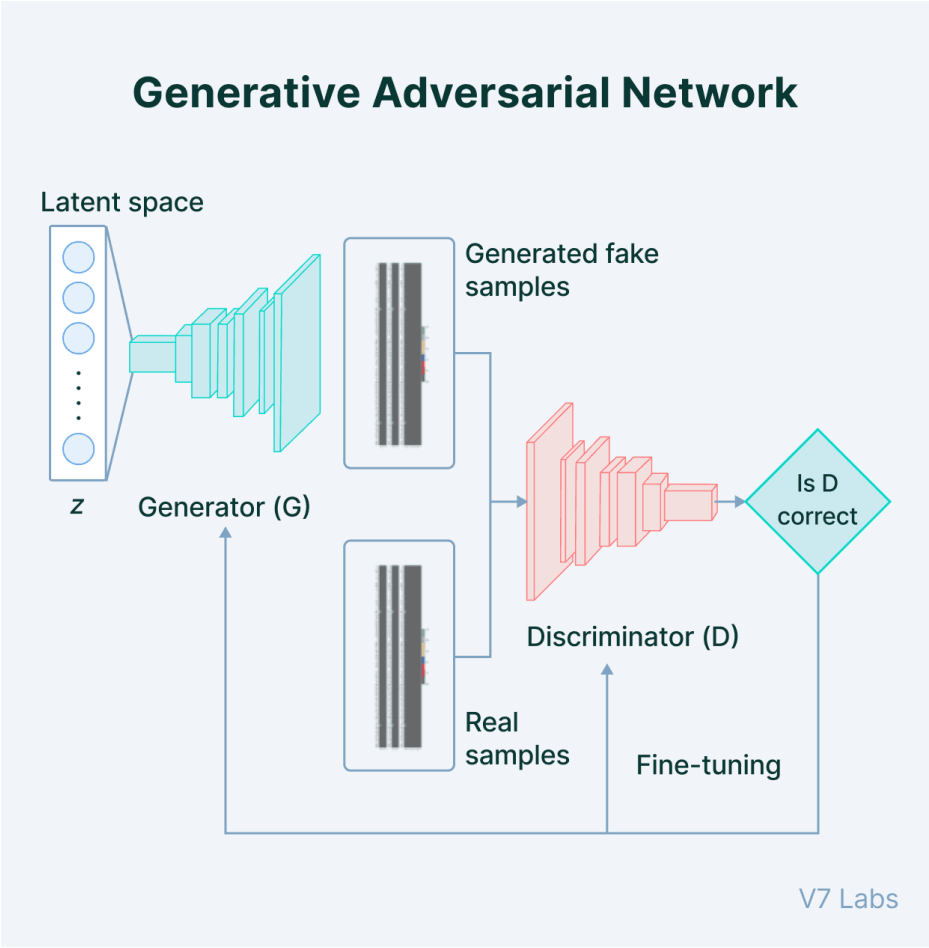
Generative AI Tools

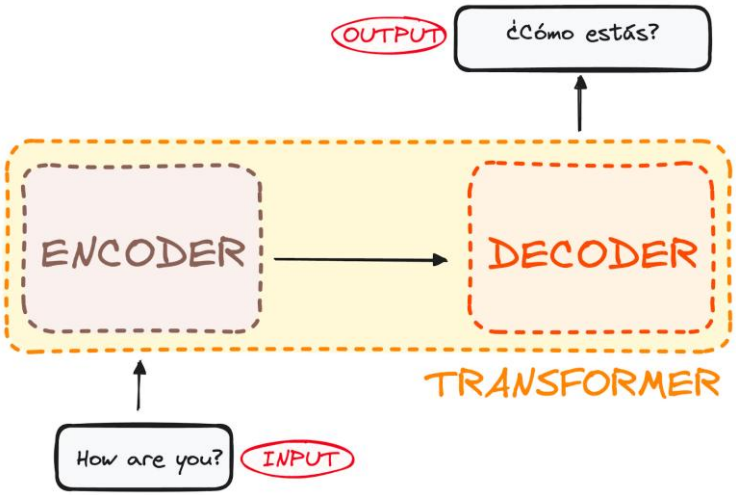
ChatGPT	• Nov. 2022 • Developed by OpenAI • https://chat.openai.com/
Copilot	• Feb. 2023 • Microsoft (powered by OpenAI) • https://www.bing.com
Gemini	• Mar. 2023 • Developed by Google • https://gemini.google.com/
Claude AI	• Mar. 2023 • Developed by Anthropic • https://claude.ai/
Others	• LLaMA (Facebook), Notion, Jasper, DALL-E, and many more.

Defining the Terminology

- **Natural Language Processing (NLP)**
  - A field of AI where computer science meets linguistics to allow computers to understand and process human language.
- **Large Language Models (LLMs)**
  - These models are built using deep learning techniques, which enable them to understand the nuances of language.
- **Generative AI**
  - A type of AI system capable of generating text, images, or other media in response to prompts,
    - e.g. using LLMs to predict and generate the most probable words in a sequence based on large data sets.

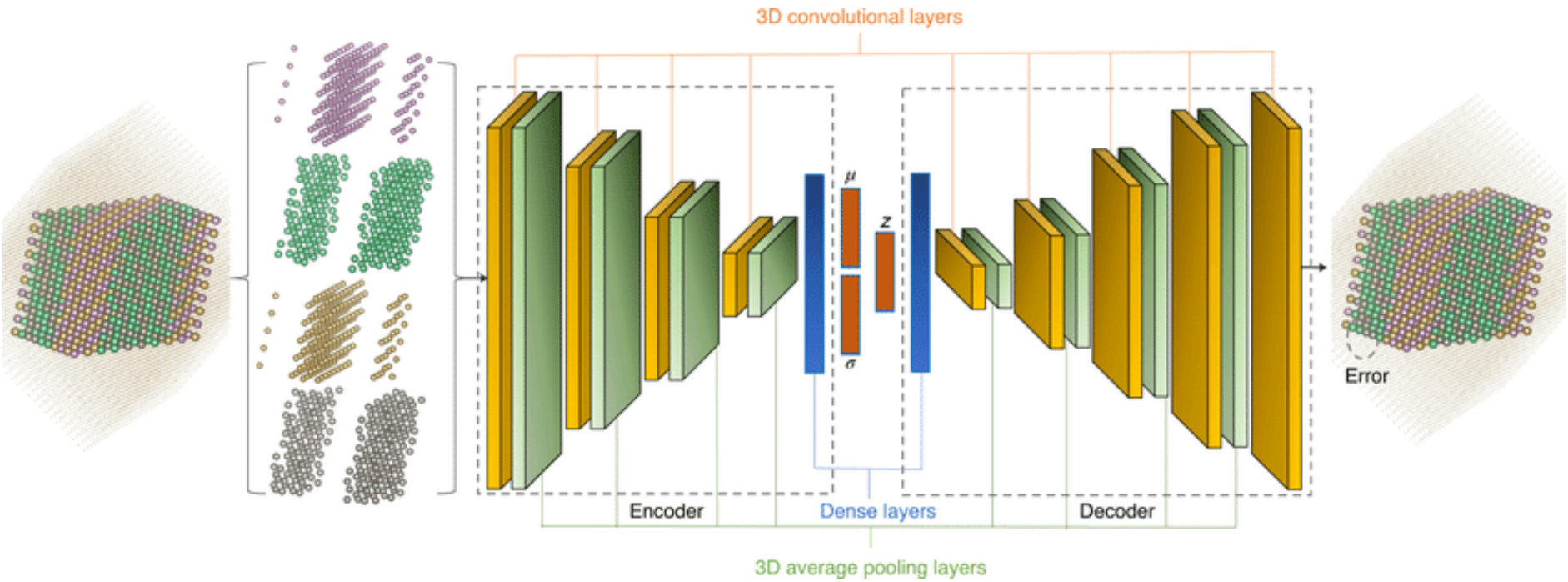
Generative Models





[The Essential Guide to Neural Network Architectures]

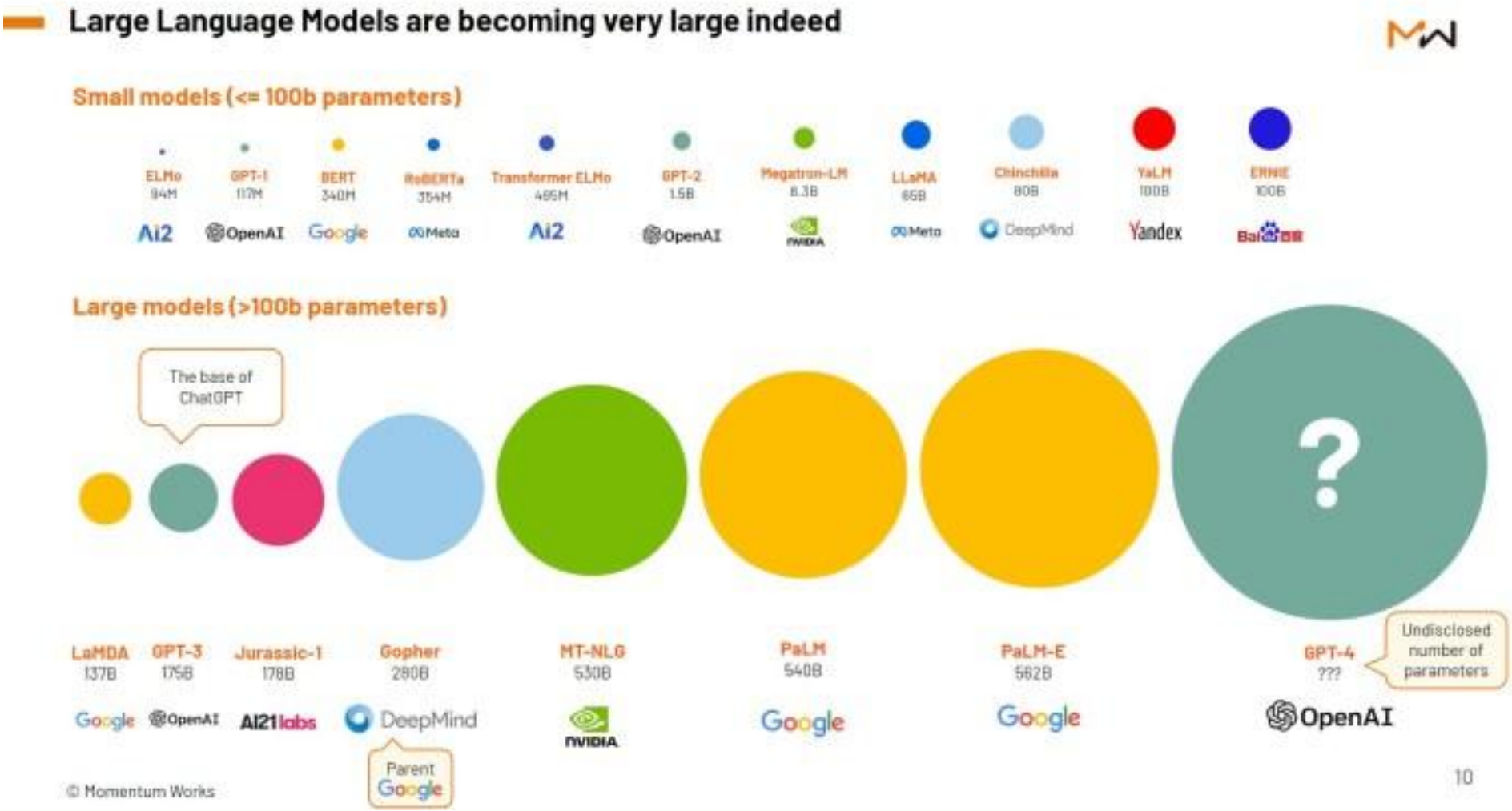
Generative Models



Variational Autoencoder (VAE)

[Neural network-based order parameter for phase transitions and its applications in high-entropy alloys]

Large Language Models (LLMs)



[Momentum Works, The emergence of Large Language Models (LLMs)]

LLMs Create Tokens

Let’s take a step back…

GPT-3 Codex

Here is a sentence broken into tokens. Notice how most words are their own token, but that sometime names or complete words like ChatGPT, or mispeeled words, and , punctuation: , "are their" own tokens! As a rule of thumb for expressions in the English language there are approximately 750 words per 1000 tokens.

Clear Show example

Tokens	Characters
68	312

Here is a sentence broken into tokens. Notice how most words are their own token, but that sometime names or complete words like ChatGPT, or mispeeled words, and , punctuation: , "are their" own tokens! As a rule of thumb for expressions in the English language there are approximately 750 words per 1000 tokens.

How Large Language Models (LLMs) are Made?

1. **Collect vast amounts of data** (training data)
2. **Transform data into tokens** (parts of words) → trained on 500 billion tokens!
3. **Unsupervised learning**: training the model (weeks to months).
   - It creates a huge mathematical representation of human language.
4. **Fine-tuning:**
   - Expose it to new tasks and give it more guidelines
5. **Reinforcement learning with human feedback (RLHF):**
   - Use human reviewers to rate responses to improve quality.
   - The model learns from those ratings.
6. **Deploy to the public**:
   - GPT 3.5 → "a research preview."
   - Learn from more people using it so it can be improved.

Basic Text Prediction

The students studied in the _____

Word (Token)	Probability
Waterpark	1%
Attic	7%
Bedroom	50%
Library	99%

How Does Generative AI Work?

- **Generative AI (Gen AI): Refers to deep-learning models that can generate high-quality text, images, and other content based on the data they are trained on. (IBM)**
- **ChatGPT: Large Language Model-based chatbot**

GPT:
Generative Pre-Trained Transformer

Generative
creates responses

Pre-Trained
The process of training a large language model on a vast amount of data before being fine-tuned for specific tasks

Transformer
A deep learning model architecture processes and generates sequences of data based on probability.

How Does Content Generation Work?

1. **User inputs a prompt or question**
   - ***Prompt***: "Write a 150-word essay on the Impressionism movement in 19th century France and its characteristics."
2. **Use an algorithm to analyze the prompt and generate a response.**
3. **Identify keywords and phrases in the prompt to understand the context of the question**
   - ***Command***: Write an essay in 150-words
   - ***Topic***: Impressionism characteristics, 19th century France
4. **Access its database of information to find relevant information related to the prompt.**
   - What does it know about writing essays? What does it know about the time, country, and characteristics of this artistic movement?
5. **Use natural language processing to generate a response that is grammatically correct and contextually relevant to its dataset.**
6. **Present the response to the user, who can choose to continue the conversation or provide additional prompts.**

Adapted and taken from Compose & Consult: https://composeandconsult.com/the-edit-effect/ai-platforms-hallucination-effect-content-generation

ChatGPT Outcome

Shared Chat • Default (GPT-3.5)

Write a 150-word essay on the Impressionism movement in 19th century France and its characteristics.

Impressionism, a revolutionary art movement that emerged in 19th century France, marked a profound shift in the way artists approached their craft. It eschewed the rigid academic traditions of the time, focusing instead on capturing fleeting moments and the play of light on everyday scenes. Impressionist painters such as Claude Monet, Pierre-Auguste Renoir, and Edgar Degas sought to convey the immediate sensory experience of their subjects.

Characterized by several key features, Impressionism embraced vibrant colors, loose brushwork, and a focus on landscapes, urban scenes, and the leisure activities of modern life. These artists often worked en plein air, outdoors, to capture the ever-changing effects of light and atmosphere. They used broken, dappled brushwork to create a sense of movement and a vivid atmosphere. The emphasis on color and light, rather than meticulous detail, set Impressionism apart from the academic art of the era.

Impressionism not only revolutionized the art world but also paved the way for subsequent artistic movements. Its influence can still be seen today, making it a lasting testament to the power of innovation in the world of art.

Continue this conversation

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Generative AI: Functions

Language and Content Generation	Information Retrieval	Language Translation	Text Summarization	Writing Assistance	Conversational Assistance
Can create content of many types, including code	Can answer most basic, non-academic questions	"Translate this sentence in English into Italian...."	"Summarize the key plot points in Bradbury's Fahrenheit 451..."	"Improve the following sentence for clarity..."	Idea generator
Unique responses	"Explain the concept of photosynthesis in a simple and clear manner..."				"Generate three ideas for a five-page essay on..."

Generative: Strengths

Writing	Learning/Studying
• Generate ideas • Brainstorm research topics • Change citation styles • Editing and proofreading • Suggestions for databases to find literature on your topic "Correct the grammar: [paragraph]" "Identify the gaps in the literature on [topic]"	• Can explain complex concepts • Summarize information • Practice second languages "Can you explain x in simple terms?" "How about using an analogy to describe it?" "Please summarize [insert book name] by [author] and provide the top 5 important themes."

Generative AI: Limitations

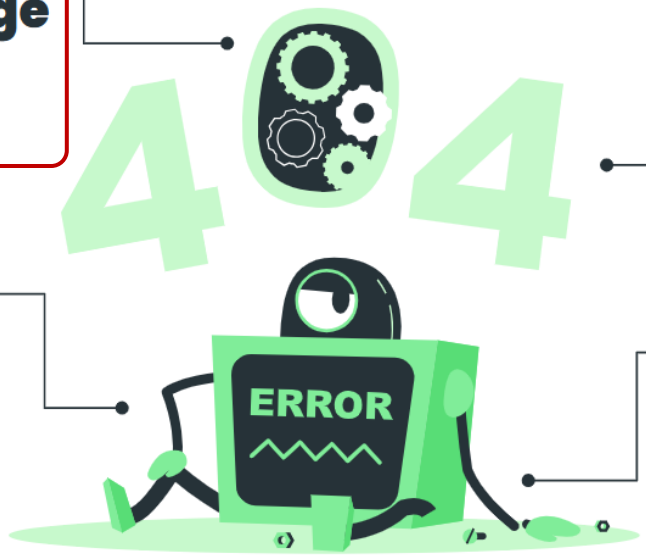
- **What are the Problems?**

Limited Knowledge

Data until January 2022,
not replicable

Bias

Misinformation



Ethical Concerns

Misinformation

- Sometimes ChatGPT makes stuff up, called hallucinations.
- **Remember:** based on the probability of what the next word should be
- Based on AI model's inherent biases and training dataset limitations.
- "*These hallucinations are particularly problematic in domains that require multi-step reasoning since a single logical error is enough to derail a much larger solution*."

Examples of Hallucinations

KU

Provide one paragraph summarizing gender-based violence and climate change with citations in APA format. Please include the reference list at the end.

Gender-based violence and climate change intersect in complex ways, exacerbating vulnerabilities and inequalities among affected populations. Climate change disproportionately impacts women and girls due to their socio-economic status and cultural roles, making them more susceptible to violence and exploitation (Abramsky et al., 2020; United Nations, 2018). Environmental disasters and resource scarcity can intensify conflicts and force displaced communities to seek alternative means of survival, exposing women to heightened risks of sexual and domestic violence (UN Women, 2019; Díaz et al., 2021). Additionally, the unequal power dynamics between genders perpetuate violence and hinder women's participation in climate change adaptation and mitigation efforts (Hossain & Scott, 2020; Naved et al., 2022). Addressing the intersectionality of gender-based violence and climate change is crucial for building resilient and equitable societies (Núñez et al., 2021).

References:

Fake References

Abramsky, T., Devries, K., Kiss, L., Nakuti, J., Kyegombe, N., Starmann, E., ... & Watts, C. (2020). Findings from the SASA! Study: A cluster randomized controlled trial to assess the impact of a community mobilization intervention to prevent violence against women and reduce HIV risk in Kampala, Uganda. BMC Medicine, 18(1), 1-15.

Díaz, S. S., Oviedo, G., & Porras, M. A. (2021). Violence against women and climate cha

Regenerate

Identify Errors & Fake Citations

1

Does it exist?

Search each reference in the library's catalogue or Google.

2

Ask for more details

"Could you provide an ISBN or ISSN for this publication?"

3

Was it properly summarized?

Consult the material to verify if it was summarized correctly.

Bias

- Bias exists in the training data itself (sexism, racism, homophobia, ableism, etc.)
- Western, English focused (developed in the US)
- Not enough information or nuance for certain types of content in the dataset
- Only humans have the skillset to think critically, conduct nuanced research, and fact-check!

Ethical Concerns

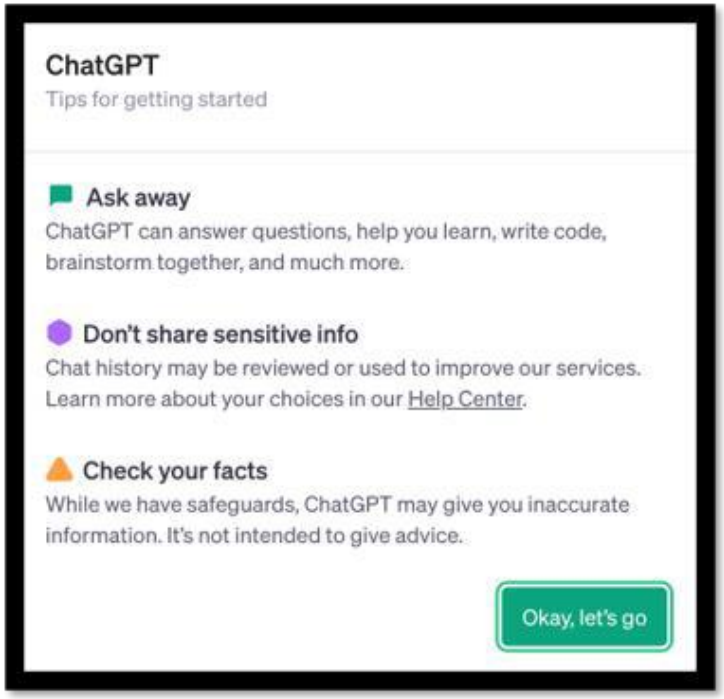
- Copyright
- Low wages to humans training ChatGPT[1]
- Dependency on AI
- What else?

BUSINESS • TECHNOLOGY

Exclusive: OpenAI Used Kenyan Workers on Less Than $2 Per Hour to Make ChatGPT Less Toxic

This image was generated by OpenAI's image-generation software, Dall-E 2. The prompt was: "A seemingly endless view of African workers at desks in front of computer screens in a printmaking style." TIME does not typically use AI-generated art to illustrate its stories, but chose to in this instance in order to draw attention to the power of OpenAI's technology and shed light on the labor that makes it possible. Image generated by Dall-E 2/OpenAI

Generative AI: Limitations



Incorrect or nonsensical responses

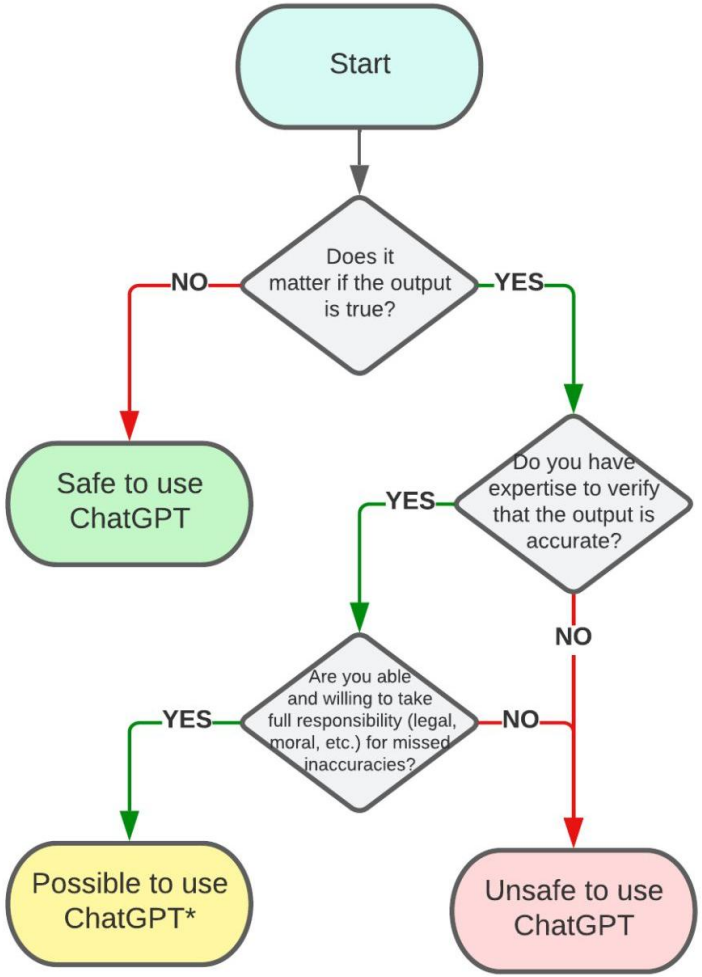
- Produces hallucinations
- Can not verify the accuracy of information (disinformation)

Over reliant on the data it is trained on: information contains biases

Does not have genuine comprehension, generates text based on patterns in data

- **"ChatGPT may produce inaccurate information about people, places, or facts." - *ChatGPT disclaimer***

Is It Safe to Use GenAI



* but be sure to verify each output word and sentence for accuracy and common sense

CC BY

Generative AI Considerations

"Attention is all you need"

arXiv:1706.03762v7 [cs.CL] 2 Aug 2023

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Generative AI Considerations

- **Lack of regulations on the technology**
  - Potential copyright issues & ownership of content
- Admin/Educators are **still working on developing policies specific** to GenAI across the nation.
- **When using GenAI, provide clear, concise directions or instructions**
  - Rephrase prompts if you are not getting a desired result
  - Use any generative AI tool with caution
  - Avoid sharing sensitive information
  - Get permission before using another's intellectual property for prompts

Generative AI Considerations

- Encourage dialogue surrounding GenAI in the classroom
- Provide your students with clear communication and expectations
- Consider ethical and legal issues when using AI tools
- Recognize that AI detection tools are not perfect
- Understand the inevitability of advancing AI technology

Exploring GEN AI In The Classroom

Bias
- Discriminatory outputs

Environment
- Mining, energy consumption, and waste

Academic Integrity
- Plagiarism, cheating

Copyright
- Infringement on intellectual property rights

Privacy
- Personal data collection

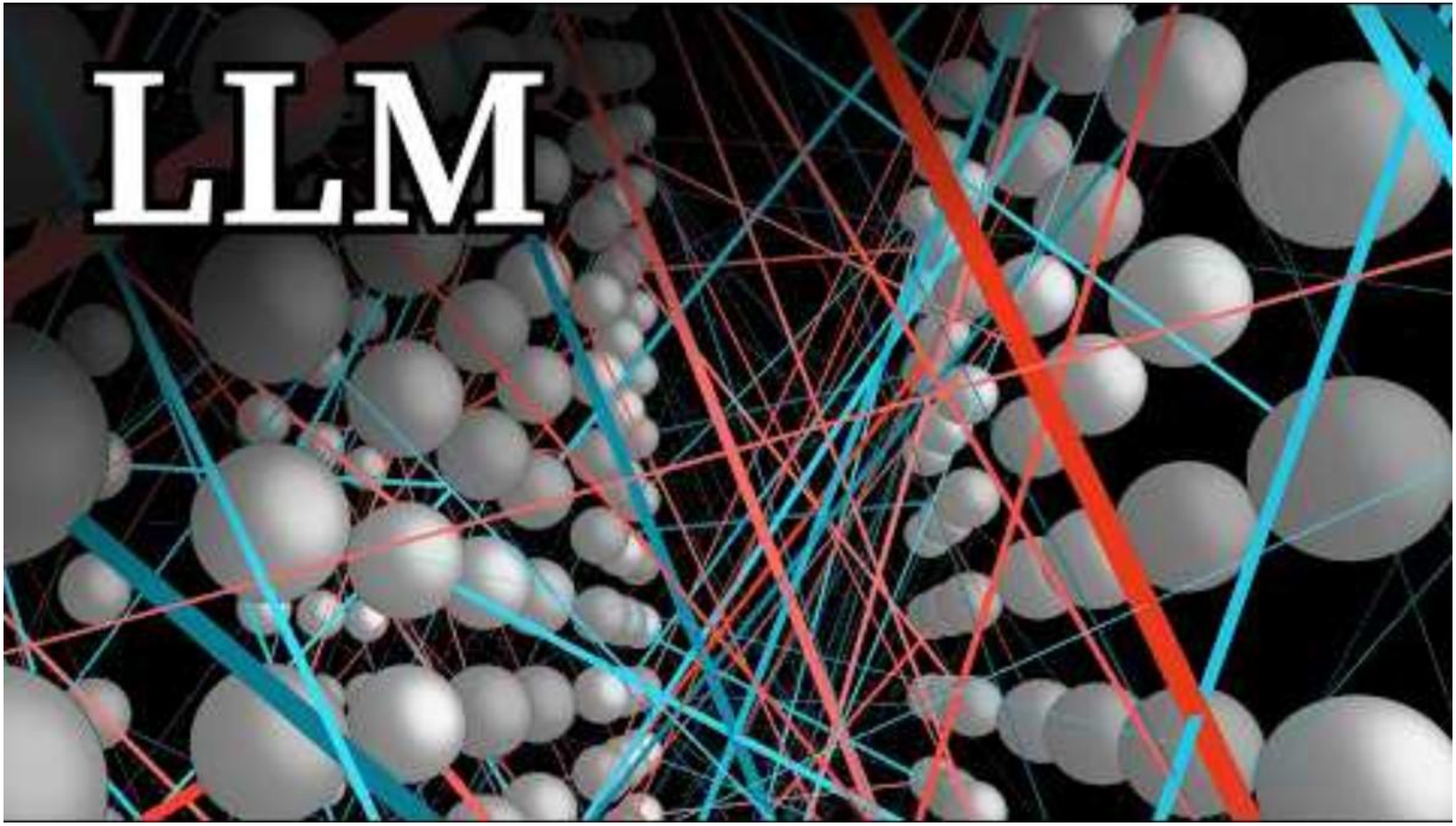
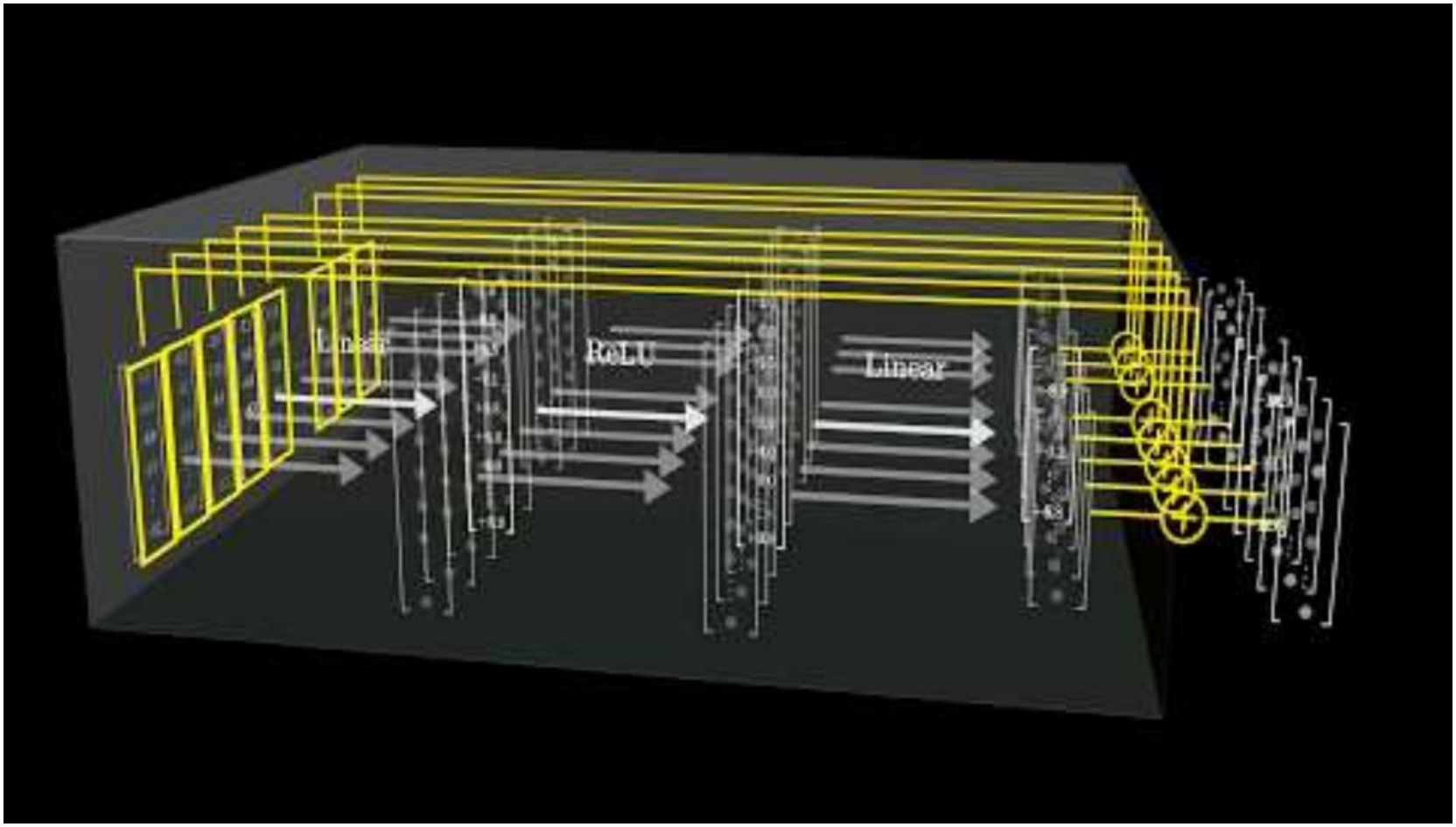
Datafication
- Commodification of personal data

Human Labor
- Job automation and exploitation

Power
- Global power imbalances, structural inequalities

Adapted and taken from Leon Furze's"Teaching AI Ethics" series: https://leonfurze.com/2023/01/26/teaching-ai-ethics/

Conclusions



Conclusions